

Program: Diploma in Food Processing Technology	
Course Code: 3421	Course Title: Food Processing Technology
Semester: 3	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:3 T:1 P: 0)	Periods per semester: 60

Course Objectives:

1. To understand the various cleaning techniques for raw materials before processing
2. To learn the concepts of drying, freezing and chilling in raw materials
3. To impart knowledge on extrusion and related concepts

Course Prerequisites:

Topic	Program/CourseName
Basic knowledge of different components of food	Fundamentals of food processing

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of raw material preparation by high temperature processing	13	Understanding
CO2	Discuss size reduction principles and equipment for solid foods	13	Understanding
CO3	Illustrate the importance of drying, refrigeration and freezing during processing	17	Understanding
CO4	Summarize the concepts of extrusion and minimal processing techniques in food technology sector.	17	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of raw material preparation by high temperature processing		
M1.01	List different types of cleaning techniques	3	Understanding
M1.02	Describe Various pasteurization methods and its importance	6	Understanding
M1.03	Explain about the blanching process	4	Understanding
Contents: Various methods of cleaning of raw materials-dry and wet techniques, pasteurization-principle, process and types with its importance- LTLT, HTST, UHT, Principle and process of blanching.			
CO2	Describe size reduction principles and equipment for solid foods		
M2.01	Explain different types of equipments used in size reduction of raw materials in food industry.	6	Understanding
M2.02	Describe the laws and principles governing size reduction	3	Understanding
M2.03	Explain the importance of homogenization	3	Understanding
	Series Test –I	1	
Contents: principle and laws of size reduction – Kicks, Bond, Rittinger’s law, Equipments used in size reduction- roller mill, impact mill, attrition mill, Importance and principle of Homogenization			

CO3	Illustrate the importance of drying, refrigeration and freezing during processing		
M3.01	Recognize different types of driers used in food processing sector	6	Understanding
M3.02	Discuss refrigeration systems and its components	6	Understanding
M3.03	Differentiate chilling and freezing in food processing	5	Understanding
Contents: Driers used in food processing industry- - Tray, tunnel, fluidized bed, spray, Rotary drier, VA,VC refrigeration systems- components - compressor, condenser, evaporator, refrigerant, COP, Chilling and freezing			
CO4	Summarize the concepts of extrusion and minimal processing techniques in food technology sector.		
M4.01	Discuss the concepts of extrusion	5	Understanding
M4.02	Explain different types of minimal processing techniques	5	Understanding
M4.03	Review hurdle technology	6	Understanding
	SeriesTest–II	1	
contents: Extrusion - Theory, operating characteristics, single screw, twin screw extruders, application, Principle, Process and applications of various minimal processing techniques- Ohmic heating, RF heating, high pressure processing, Hurdle technology - Principle and application.			

Text/Reference

T/R	Book Title/Author
R1	James G Brennan, Alistair S Grandison (Ed.), Food processing Handbook, Wiley – VCH, 2011, ISBN-13: 978-3527324682
R2	P G Smith, Introduction to Food Process Engineering, Springer, 2011, ISBN-13: 978-0306473975
R3	Zeki Berk, Food process engineering and technology, Elsevier, 2013, ISBN-13: 978-0128120187
R4	Theodoros Varzakas and Constantina Tzia, Food Engineering Handbook: Food Process Engineering, CRC Press; 1 edition (2014), ISBN-13: 978-1482261660
R5	Theodoros Varzakas and Constantina Tzia, Food Engineering Handbook: Food Process Engineering, CRC Press; 1 edition (2014), ISBN-13: 978-1482261660

R6	Saravacos G D, Food Process Engineering Operations, Taylor and Francis, 2011, ISBN: 9781420083538
R7	Toledo R T, Fundamentals of Food Process Engineering, Springer, Fourth Edition (2018), ISBN: 9783319900971

Online Resources

Sl.No	Website Link
1	https://www.millrocktech.com/lyosight/lyobrary/what-is-freeze-drying/
2	https://www.dksh.com/global-en/lab-solutions/insights/importance-of-hurdle-technology-in-food-safety
3	https://www.thepharmajournal.com/archives/2020/vol9issue2/PartD/9-1-59-634.pdf
4	https://www.vedantu.com/chemistry/homogenization